

Foundations of Artificial Intelligence

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Exercise Sheet 1 — Solutions

Exercise 1.1 (1+1+1+1 marks)

Characterize the following AI systems with respect to the four categories (acting humanly, thinking humanly, thinking rationally, acting rationally) that have been introduced in the lecture. Justify your answers in one to two sentences.

- (a) Pepper (https://en.wikipedia.org/wiki/Pepper_%28robot%29)
- (b) AlphaGo (<https://en.wikipedia.org/wiki/AlphaGo>)
- (c) ACT-R (<https://en.wikipedia.org/wiki/ACT-R>)
- (d) Google Maps (specifically for route planning; https://en.wikipedia.org/wiki/Google_%28Maps%29)

Note: In many cases several categories can be argued for, meaning there is not only one correct answer. Choose one category and present your argument for it.

Solution:

- (a) Pepper is primarily used for tasks which require social interaction, such as receptionist or caregiver to the elderly. For such tasks it is desirable to *act humanly*, and his ability to read emotions is helpful in this regard.
- (b) AlphaGo became proficient in playing Go by playing against itself millions of times, which is clearly not mimicking human behavior, making it an example of *acting rationally*.
- (c) ACT-R is a cognitive architecture used to develop models of human cognition. Despite having “act” in its name and the R standing for “rational”, it fits best in the *think humanly* category. The goal of a cognitive model is to represent human cognition, including its flaws. For example, tasks that are difficult for a human should take longer than tasks that are easy for a human. Forgetting, confusion, reaction times, etc. can be modelled in ACT-R.
- (d) Google Maps uses geo data, road networks and live traffic data for route planning. Based on this data, it recommends a route between a given pair of locations which minimizes travel time and/or distance based on the users’ preferences (e.g., mode of transport). Because we are interested in the outcome rather than the internal algorithms of the system, we categorize this as *acting rationally*.

Exercise 1.2 (1.5+1.5 marks)

- (a) Consider a self-driving taxi that takes the most scenic route instead of the fastest route. Does it behave rationally? Justify your answer.
- (b) Consider ChatGPT, a chatbot designed to be a digital assistant and answer questions. There are various examples where we find that ChatGPT provides incorrect answers and tricks users by trying to make the answer sound correct. Is this behavior rational?
Hint: ChatGPT is a language model designed to predict the most likely next words in a series of words and it was trained with a form of reinforcement learning that incorporates human feedback (see RLHF¹).

¹<https://openai.com/blog/deep-reinforcement-learning-from-human-preferences/>

Solution:

- (a) It depends on the performance measure of the taxi which can for example be influenced by the passenger. If the goal is to reach the destination as fast as possible then this behaviour would not be rational. If however the goal is to pass by as many beautiful locations as possible, then the taxi behaves rationally according to its performance measure.

- (b) Yes, this behavior is rational, since the goal is to provide answers which sound correct to a human but don't necessarily need to be correct. As long as the human is satisfied with the answer, the correctness is irrelevant.

The AI was trained to receive positive feedback from the human evaluators, i.e. it doesn't matter if they themselves don't know the correct answers since humans have tendencies to rate incorrect answers that sound correct higher than the answer "I don't know the answer". This leads to the AI learning what the human evaluators think is right, i.e., the AI learns to trick the human evaluators by giving them the most plausible answer rather than giving them the actual correct results.

Exercise 1.3 (1.5+1.5 marks)

Consider an agent in a 2D grid where certain cells are blocked trying to reach the upper right corner of the grid. The agent can observe whether it is at the goal or not, and whether the cells north, south, east and west are blocked.

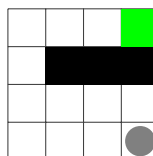
- (a) Provide a description of a possible implementation for such a reflexive agent.
- (b) Is the agent guaranteed to find the goal (if it is reachable) in finitely many steps? Justify your answer.

Solution:

- (a) The following algorithm is one possible implementation:

```
while Not at goal do
  if North is free then
    | Go North
  else if East is free then
    | Go East
  else if South is free then
    | Go South
  else if West is free then
    | Go West
  else
    | Give up
  end
end
```

- (b) No, the agent may get stuck. Consider the following grid with the agent (gray circle) in the lower right corner:



The agent will first go north, and then iterate between going south and north, never reaching the goal (in green).